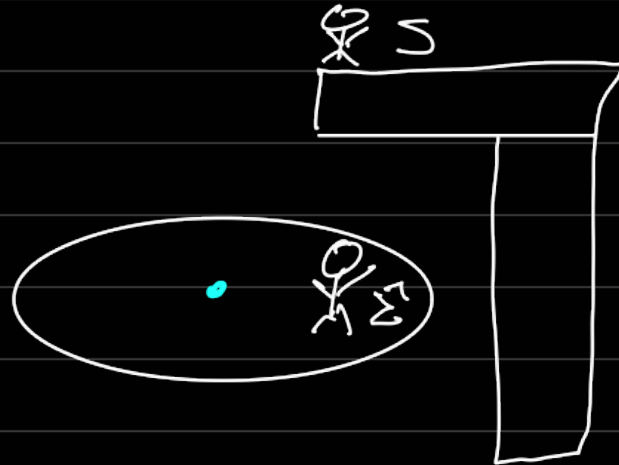


* Recall $\frac{d\phi}{dt} = \omega(t),$

$$\frac{d\vec{r}}{dt} = \vec{\omega} \times \vec{r}$$

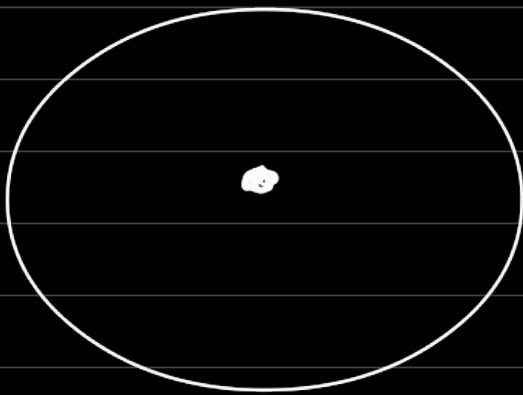
* Problem set up:

- (i) stationary reference frame we label by S
- (ii) Moving reference frame we label by Σ

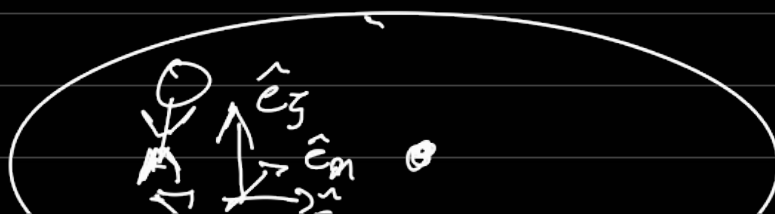


- For both reference frames we call the center of the circle the origin

S : We imagine someone looking over the disk so a person at S sees



Σ : Here this is a person on the disk



$$\vec{r}_s = x(t) \hat{e}_x + y(t) \hat{e}_y + z(t) \hat{e}_z$$

$$\vec{r}_\Sigma = \xi(t) \hat{e}_\xi(t) + \eta(t) \hat{e}_\eta(t) + \zeta(t) \hat{e}_\zeta(t)$$

↑ Here the person has a basis $(\hat{e}_\xi, \hat{e}_\eta, \hat{e}_\zeta)$ which is also moving with the disk as it spins

• The two expressions should be the same

$$\vec{r}_s = \vec{r}_\Sigma$$

since they both use the same origin

→ They differ once we start looking at velocities and accelerations

* In my last notes we found

$$\underline{\vec{a}_s = \vec{a}_{\Sigma, \text{app}} + \dot{\vec{\omega}} \times \vec{r}_{\Sigma}(t) + \vec{\omega} \times (\vec{\omega} \times \vec{r}_{\Sigma}) + 2\vec{\omega} \times \vec{v}_{\Sigma, \text{app}}}$$

i) Since ω is constant, then $\dot{\vec{\omega}} = 0$

so

$$\dot{\vec{\omega}} \times \vec{r}_\Sigma = 0$$

ii) $\vec{a}_s$ is the acceleration as seen by a person in S .

But

The problem gives us $\vec{r}_s$

$$\vec{r}_s = (0, -Ut)$$

So its velocity is

$$\vec{v}_s = \dot{\vec{r}} = (0, -U)$$

and the acceleration is

$$\vec{a} = \frac{d^2 \vec{r}_s}{dt^2} = (0, 0)$$

hence $\vec{a} = 0$

→ Plugging in $\vec{a}$ and $\dot{\omega} = 0$ we are left with

$$\vec{a}_{s,app} = -\vec{\omega} \times (\vec{\omega} \times \vec{r}_s) - 2\vec{\omega} \times \vec{v}_{s,app}$$

Recall triple scalar product

$$\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b})$$

$$\vec{a} = \vec{\omega}$$

$$\vec{b} = \vec{\omega}$$

$$\vec{c} = \vec{r}_s$$

$$\vec{\omega} \times (\vec{\omega} \times \vec{r}_s) = \vec{\omega}(\vec{\omega} \cdot \vec{r}_s) - \vec{r}_s \vec{\omega} \cdot \vec{\omega}$$

$$\vec{\omega} \cdot \vec{r}_s = \omega_x \xi + \omega_y \eta + \omega_z \zeta = 0$$

• why is this zero? Well

$$\vec{\omega} = \omega \hat{e}_z = \underbrace{0}_{\omega_x} \hat{e}_x + \underbrace{0}_{\omega_y} \hat{e}_y + \underbrace{\omega}_{\omega_z} \hat{e}_z$$

So $\omega_x = \omega_y = 0$ and the first two terms vanish.

• Since $\vec{r}_s$ has no z-component (it lies parallel to the disk)

$$\vec{r}_s \cdot \hat{e}_z = 0 \quad (\zeta = 0)$$

hence last term is zero. So we are left with

$$\vec{\omega} \times (\vec{\omega} \times \vec{r}_s) = -\vec{r}_s \omega^2$$

Plugging this back in

$$\vec{a}_{s,app} = -\vec{r}_s \omega^2 - 2\vec{\omega} \times \vec{v}_{s,app}$$

$$\vec{a}_{S,app} = \vec{\omega} \times (\vec{\omega} \times \vec{r}_{S,app}) - 2 \vec{\omega} \times \vec{v}_{S,app}$$

- So we're almost there, let's do the second term

$$2 \vec{\omega} \times \vec{v}_{S,app}$$

• Recall:

$$\vec{v}_{S,app} = \dot{\xi} \hat{e}_{\xi} + \dot{\eta} \hat{e}_{\eta}$$

$$\vec{\omega} \times \vec{v}_{S,app} = \omega \hat{e}_z \times (\dot{\xi} \hat{e}_{\xi} + \dot{\eta} \hat{e}_{\eta})$$

$$= \omega \dot{\xi} (\hat{e}_z \times \hat{e}_{\xi}) + \omega \dot{\eta} (\hat{e}_z \times \hat{e}_{\eta})$$

- Now use the relationships given to us in the problem:

$$\hat{e}_{\xi} \times \hat{e}_{\eta} = \hat{e}_z$$

$$\hat{e}_{\eta} \times \hat{e}_z = \hat{e}_{\xi} = -\hat{e}_z \times \hat{e}_{\eta}$$

$$\hat{e}_z \times \hat{e}_{\xi} = \hat{e}_{\eta}$$

So

$$\vec{\omega} \times \vec{v}_{S,app} = \omega \dot{\xi} \hat{e}_{\eta} + \omega \dot{\eta} (-\hat{e}_{\xi})$$

* Now we plug the above in

$$\vec{a}_{\Sigma, \text{app}} = \vec{r}_{\Sigma} \omega^2 - 2\omega (\dot{\xi} \hat{e}_n - \dot{\eta} \hat{e}_{\xi})$$

(2) Set up: I'll set this up and you should try and solve the rest

$$\vec{a}_{\Sigma, \text{app}} = \ddot{\vec{r}}_{\Sigma, \text{app}} = \ddot{\xi}(t) \hat{e}_{\xi} + \ddot{\eta} \hat{e}_n \quad (\text{LHS})$$

$$\begin{aligned} (\text{RHS}) &= \vec{r}_{\Sigma} \omega^2 - 2\omega (\dot{\xi} \hat{e}_n - \dot{\eta} \hat{e}_{\xi}) \\ &= (\xi \hat{e}_{\xi} + \eta \hat{e}_n) \omega^2 - 2\omega (\dot{\xi} \hat{e}_n - \dot{\eta} \hat{e}_{\xi}) \end{aligned}$$

• collect basis terms

$$= (\xi \omega^2 + 2\omega \dot{\eta}) \hat{e}_{\xi} + (\eta \omega^2 - 2\omega \dot{\xi}) \hat{e}_n$$

Now we equate the vector components on the (LHS) & (RHS), this gives us

$$\ddot{\xi} = \xi \omega^2 + 2\omega \dot{\eta} \quad \text{and} \quad \ddot{\eta} = \eta \omega^2 - 2\omega \dot{\xi}$$

* Now use complex coordinates:

$$Z(t) \equiv \xi(t) + i\eta(t)$$

$$\dot{Z}(t) = \dot{\xi} + i\dot{\eta}$$

$$\ddot{Z}(t) = \ddot{\xi} + i\ddot{\eta}$$

• Plug in $\ddot{\xi}$ and $\ddot{\eta}$ from above

$$\begin{aligned} \ddot{Z} &= (\xi \omega^2 + 2\omega \dot{\eta}) + i(\eta \omega^2 - 2\omega \dot{\xi}) \\ &= \omega^2 (\underbrace{\xi + i\eta}_Z) + 2\omega (\underbrace{\dot{\eta} - i\dot{\xi}}_{-i\dot{Z}}) \end{aligned}$$

Thus

$$\ddot{Z} = \omega^2 Z - 2i\omega \dot{Z}$$

$$z = \omega^{-1} \ddot{z} - 2i\omega \dot{z}$$

or $\ddot{z} + 2i\omega \dot{z} - \omega^2 z = 0$

as claimed in the problem. Try to solve this to obtain ξ and η as given.

